JAGAT GURU NANAK DEV PUNJAB STATE OPEN UNIVERSITY, PATIALA

DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS

M.Sc. (Computer Science) — Semester-IV

M.Sc. (Computer Science) Semester-IV Master Portfolio

Jagat Guru Nanak Dev Punjab State Open University, Patiala (JGND PSOU)

Academic Year: 2026 | Degree: Master of Science in Computer Science (M.Sc. CS)

📌 University Course Inventory & Marks Matrix

Course Code	Course Name	Credits	External Marks	Internal Marks	Total Mark
MSCS-4-03P	Major Project	10	140	60	200
MSCS-4-04	Seminar	2	0	50	50
TOTAL	Final Semester Portfolio	12	140	110	250

🏛️ Project & Research Alignment (SSR ONE AI)

Both courses are systematically anchored in the enterprise codebase `SSR One AI` (`e:/2026/ssr_one_ai`), an advanced multi-tenant hospitality SaaS POS and ERP system.

Major Project (MSCS-4-03P):

Approved Title: *"Design and Implementation of a High-Throughput, Offline-First, Multi-Tenant SaaS Enterprise POS Architecture with Integrated AI Copilot and Computer Vision OCR"*

Scope: Full-stack enterprise architecture encompassing React 19, Dexie.js IndexedDB, FastAPI ASGI, PostgreSQL 16 Row-Level Security (RLS), WebSockets, and Tesseract OCR.

Seminar (MSCS-4-04):

Approved Topic: *"Architectural Patterns for Offline-First Distributed Web Applications: Balancing Zero-Latency User Experience with Eventual Consistency"*

Scope: Theoretical and practical research on local-first web storage, asynchronous reconciliation queues, idempotency tokens, and distributed conflict resolution.

📂 Repository Directory Structure (`msccs/`)

```
e:/2026/ssr_one_ai/msccs/
├── 00-MASTER-INDEX.md                          # Master Portfolio & University Guideli
│
├── 01-MAJOR-PROJECT-MSCS-4-03P/                # Course Code: MSCS-4-03P (200 Marks)
│   ├── 01-TOPIC-APPROVAL-APPLICATION.md       # Formal Topic Application to Course Coo
│   ├── 02-PROJECT-SYNOPSIS.md                  # University-Mandated 8-Part Synopsis (
│   ├── 03-FINAL-PROJECT-REPORT-THESIS.md       # Complete 30-35 Page Final Project Rep
│   ├── 04-PRESENTATION-SLIDES.md               # 10-15 Min Viva Presentation Deck & Sc
│   ├── 05-VIVA-VOCE-DEFENSE-GUIDE.md           # Comprehensive External Examiner Q&A G
│   └── 06-RESEARCH-PAPER-IEEE.md               # Academic Research Paper Draft (IEEE 2
│
└── 02-SEMINAR-MSCS-4-04/                       # Course Code: MSCS-4-04 (50 Marks)
    ├── 01-SEMINAR-TOPIC-APPROVAL-APPLICATION.md # Formal Seminar Topic Application Le
    ├── 02-SEMINAR-REPORT.md                    # 10-15 Page Academic Seminar Report (3
    ├── 03-SEMINAR-PRESENTATION-SLIDES.md       # 15-20 Min Oral Presentation Slide Dec
    └── 04-SEMINAR-VIVA-QA.md                   # Question Handling & Discussion Guide
```

📅 University Deadlines & Submission Milestones

Milestone	University Window	Document to Submit	Target Deadline
Topic Approval	Beginning of 4th Semester	Topic Application Letters (Project + Seminar)	Week 1
Project Synopsis	Within 15 Days of Approval	`02-PROJECT-SYNOPSIS.md` (8-Part Format)	Week 3
Seminar Report	Within 30 Days of Approval	`02-SEMINAR-REPORT.md` (10–15 A4 Pages)	Week 5
Seminar Presentation	Scheduled by Coordinator	`03-SEMINAR-PRESENTATION-SLIDES.md` (15–20 Min)	Week 6
Final Project Report	At least 1 week before Viva	`03-FINAL-PROJECT-REPORT-THESIS.md` (30–35 Pages)	Week 10
Major Project Viva	End of Semester	Presentation (10–15 Min) + Viva Discussion (7 Min)	Week 12
Paper	Semester Concluding	`06-RESEARCH-PAPER-`	Post-Viva